

Picking a fraud model is a trade-off, not a leaderboard

To: Head of Fraud Risk · **From:** Amina Khaknazarova · **Date:** October 2026

This memo is based on an MSc dissertation at Loughborough University London. It uses a public card fraud dataset of 284,807 transactions (492 fraud cases), reduced to 283,726 transactions (473 fraud cases) after removing 1,081 duplicates. The data is not from a UK bank and the currency is unconfirmed, so amounts are given in monetary units.

Recommendation

Do not choose a fraud model on AUC alone. Logistic Regression, Random Forest and XGBoost showed no statistically significant difference in detection, although with only five cross-validation folds the test has limited power. The models differ mainly in how easily they can be validated and explained. Model choice should therefore be recorded as a trade-off: weaker detection with an easier compliance case (Logistic Regression), or stronger detection with a harder one (Random Forest or XGBoost).

Findings

- **Detection.** On the held-out test set (56,746 transactions, 95 fraud cases), AUC ranged from 0.956 to 0.975 across the three models. In five-fold cross-validation, the differences were not significant ($p = 0.29$ to 0.92).
- **Calibration.** Training on a balanced sample distorts predicted probabilities. The Dal Pozzolo et al. (2015) correction reduced the Brier score by 95.0% to 96.8% for all three models.
- **Cost.** Expected loss was calculated as the corrected XGBoost probability multiplied by the transaction amount. It totalled 34,217.60 monetary units on the test set, of which 72.7% came from false alarms and 26.1% from fraud that was caught. A single genuine transaction of 25,691.16, flagged at a probability of 68.6%, accounted for 51.5% of the total.

Risks

- **False alarms.** At the default 0.5 threshold, precision is 4.8% to 6.8% at 85% to 87% recall, which means 1,113 to 1,647 false alarms for 81 to 83 frauds caught. The threshold was not derived from costs.

- **Concentration.** A count of false positives would have hidden that one customer accounted for half of the estimated loss. Concentration should be tracked alongside volume.
- **Explanations.** SHAP was stable: V14 was the top feature in all 5 Random Forest folds and 4 of 5 XGBoost folds. LIME was consistent across 20 reruns (Jaccard 0.694 for XGBoost, 0.771 for Random Forest) but fitted the model poorly (R^2 0.365 to 0.388). The leading features are also anonymised components, so a blocked customer could not be given a meaningful reason.

UK regulatory fit

Provision	Result
Consumer Duty, foreseeable harm	Well supported: false alarms dominate cost
DUAA 2025 22C(a), information to customer	Technically satisfiable, practically weak
DUAA 2025 22C(d), ability to contest	Partially satisfied: stable but low fit
PRA SS1/23 Principle 3, documentation	Satisfied in principle
PRA SS1/23 Principle 4, validation	Quantified tension: best-performing model is hardest to validate

Six further provisions (22C(b) and (c), the good-faith rule, and SS1/23 Principles 1, 2 and 5) depend on how a firm operates rather than on model output, so they could not be tested.

Recommended actions

1. Record the model choice as a risk decision at deployment.
2. Track the concentration of false positives, not just their count.
3. Audit explanations for both stability and local fit before relying on LIME for compliance.
4. Treat SS1/23 validation effort as a real cost when comparing models.
5. Repeat the analysis on UK data, with a cost-based threshold, before generalising.

Limits: the data is European, default hyperparameters were used, and the absence of demographic attributes means fairness could not be tested. With only 95 fraud cases in the test set, the fraud metrics could shift on a different split.